

Explainable Cyberbullying Detection in Hinglish: A Generative Approach

Krishanu Maity , Raghav Jain , *Member, IEEE*, Prince Jha , and Sriparna Saha 

Abstract—The escalating prevalence of online cyberbullying and trolling across various social media platforms has become a pressing concern. Extensive research demonstrates the detrimental impact of cyberbullying on the mental well-being of its victims. Given the sheer volume of online content, manual identification of cyberbullying instances proves unfeasible, necessitating the development of automated cyberbullying detection methods. This challenge has attracted considerable attention within the natural language processing (NLP) community, owing to advancements in machine learning techniques. However, most of the methods fail to provide reasoning for their decisions which warrants the use of interpretable models that can explain the model’s output in real-time. Interpretable models rather than black-box models with high performance are becoming popular adhering the “right to explanations” laws. Motivated by this, we create a cyberbullying corpus *BullyExplain* in code-mixed language, where a post has been annotated with four labels, i.e., bully, sentiment, target, and rationales (explainability). Current work addresses the task of explainable cyberbullying detection and proposes a unified generative framework, *BullyGen* by redefining this multitask problem as a text-to-text generation task. Our framework is capable of not only detecting whether the text is a cyberbully or not but also provides reasoning by predicting rationale, target group and sentiment of the text. Experimental results illustrate the efficacy of our proposed model by outperforming the state-of-the-art and several baselines by a significant margin and conclude that text-to-text generation model could be a good alternative for multitask classification problems.

Index Terms—Code-mixed, cyberbullying detection, explainability, multitask, sentiment, target.

I. INTRODUCTION

THE rise of social media platforms has profoundly reshaped our communication patterns, ushering in a concerning surge in improper behaviors, including the use of offensive language to target individuals. This phenomenon, known as cyberbullying [1], involves electronic communication that denigrates individuals based on personal attributes such as gender, ethnicity, and more. According to the Pew Research Center, 40% of social media users have reported encountering online harassment [2]. Victims of cyberbullying often experience

emotional distress and lowered self-esteem [3], necessitating the development of automatic detection techniques with explainable models to mitigate these adverse effects.

Though significant research has focused on automatic cyberbullying detection using traditional deep learning [4], [5], [6], [7] and machine learning approaches [8], [9], [10], [11], the escalating use of code-mixed language in diverse multilingual societies presents new challenges [12]. This linguistic blending, termed code mixing [13], reflects the complex interplay of multiple languages within communication. While code-mixed offensive content detection, including hate speech [14] and cyberbullying [15], has gained traction in the natural language processing (NLP) community, a critical aspect, explainability, remains underexplored. The ascent of explainable artificial intelligence (AI) [16] has underscored the need for transparent decision-making by machine learning systems, reinforced by regulations like the General Data Protection Regulation (GDPR) [17] and its “right to explanation” provision. Thus, the imperative lies in crafting interpretable models that balance performance with transparency, enabling trust, and comprehension in AI applications.

To address this research gap, we introduce *BullyExplain*, a novel explainable cyberbullying dataset tailored to code-mixed language. It emerges from a preexisting Hindi–English cyberbullying dataset, augmented with bully and sentiment labels, and subsequently enriched with target categories and rationales (explainability). In this dataset, we highlight certain phrases or words (rationales) important for identifying a post as a bully. When models use these rationales, they become better at making decisions similar to humans, making the process more understandable and trustworthy. Conversely, nonbully posts lack target categories and rationales.

Multitask learning (MTL) [18] is a training paradigm where models adeptly handle related tasks, fostering efficient learning and connection mapping. The conventional MTL model employs a shared encoder featuring task-specific layers, yet this approach faces pitfalls like negative transfer [19], limited model capacity [20], or optimization scheme [20] (how to assign weights to different tasks during training). Moreover, scalability woes arise, requiring adjustments for every new task addition or task combination. To surmount these MTL challenges, we propose a unified generative framework named *BullyGen* for solving the complex task of explainable cyberbully detection. In this text-to-text generation framework, cyberbully detection (CD) and rationale detection (RD) form primary tasks, while target identification (TI) and sentiment analysis (SA)

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The authors are with the Department of Computer Science and Engineering, Indian Institute of Technology Patna, Patna 801103, India (e-mail: krishanu_2021cs19@iitp.ac.in; raghavjain106@gmail.com; princekumar_1901cs42@iitp.ac.in; sriparna@iitp.ac.in).

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serve as secondary tasks. *BullyGen* pioneers a correction mechanism (CM) to effectively manage the code-mixed dataset's noise. Extensive experiments underscore *BullyGen*'s superiority over baselines and state-of-the-art models across various evaluation metrics.

Here are the summaries of our contributions.

- 1) We explore two issues: a) the development of an interpretable system for detecting cyberbullying in code-mixed text; and b) addressing multitasking as a text-to-text generation problem.
- 2) We have curated a dataset named *BullyExplain* specifically designed for cyberbullying detection, incorporating TI within the code-mixed context.
- 3) Introducing a unified generative framework, denoted as *BullyGen*, which addresses four concurrent tasks: a) bully detection; b) SA; c) RD; and d) TI.
- 4) Through experimental results, we demonstrate that incorporating rationale, sentiment, and target information significantly improves the efficacy of cyberbullying detection.

II. RELATED WORKS

The realm of text mining and NLP has been extensively applied to investigate diverse facets of cyberbullying detection. Prior research has delved into an array of pertinent domains, including the identification of online sexual predators, the recognition of internet abuse, and the detection of cyberterrorism [21]. This section presents a comprehensive review of recent developments in both English and code-mixed data scenarios, as well as the incorporation of rationales to enhance model interpretability.

A. Works on English Data

Badjatiya et al. [7] undertook extensive experimentation involving a range of deep learning architectures to acquire semantic word embeddings. Their exploration, conducted on a benchmark dataset comprising 16 000 annotated tweets, unveiled the superiority of deep learning approaches, surpassing state-of-the-art char/word n-gram algorithms by an impressive 18% in terms of F1 score. The study also revealed that the fusion of deep neural network embeddings with gradient-boosted decision trees yielded the highest levels of accuracy. In a different avenue, a study by Dinakar et al. [9] delved into cyberbullying detection by analyzing a corpus of 4500 YouTube comments, employing diverse binary and multiclass classifiers. Their investigation showcased that the SVM classifier achieved an overall accuracy of 66.70%, outperforming the Naive Bayes classifier. The researchers emphasized the potential of developing topic-sensitive classifiers to mitigate textual cyberbullying. Turning to another approach, Reynolds et al. [10] harnessed data from Formspring.me to establish a cyberbullying dataset, achieving a commendable 78.5% accuracy utilizing the C4.5 decision tree method. Meanwhile, Balakrishnan et al. [22] introduced a cyberbullying detection algorithm in 2020, combining various machine learning methodologies with psychological attributes

of Twitter users. Their study revealed that integrating personality and sentiment traits with baseline data (text, user, and network) significantly enhanced the accuracy of cyberbullying detection. Furthermore, the advent of BERT-based approaches marked a significant stride in cyberbullying detection. The CyberBERT system, devised by [23], demonstrated cutting-edge performance across benchmark cyberbullying datasets from Formspring (12 000 posts), Twitter (16 000 posts), and Wikipedia. Notably, the proposed model's performance outshone existing attention-based deep neural network models, marking a pivotal advancement in the field. Obaid et al. [5] developed a technique to assess the severity of cyberbullying using deep learning and fuzzy logic, analyzing 47 733 Twitter comments to identify and categorize cyberbullying.

B. Works on Code-Mixed Data

Maity et al. [24] introduced a novel BERT-Capsule model to detect cyberbullying within code-mixed language contexts. Their model underwent training on a dataset comprising tweets in both Hindi and English, each meticulously annotated as either indicative of bullying or not. Notably, the proposed model achieved an impressive accuracy rate of 79.28. In the pursuit of refining language-based models for identifying toxic content, a deep learning-driven, domain-specific word embedding model highlighted in [25] demonstrated remarkable prowess. Operating within the realm of Hindi-English code-mixed data, this model exhibited a notable superiority of 12% over conventional machine learning classifiers in effectively recognizing instances of hate speech. This enhancement in precision marked a significant advancement in addressing language complexities. Further enriching the code-mixed language domain, Kumar et al. [26] embarked on developing an annotated corpus focused on the spectrum of aggressiveness. Encompassing a substantial 21 000 Facebook comments and 18 000 tweets written in the intricate realm of Hindi-English code mixing, this dataset holds immense promise. Beyond cyberbullying, this resource also holds the potential to automatically identify related phenomena, such as trolling, prevalent across the internet landscape, particularly within social media platforms. Maity et al. [6] introduce *BullySentEmo*, a benchmark Hindi-English code-mixed corpus for cyberbullying detection, incorporating sentiment and emoji information.

C. Works on Rationales

Zaidan et al. [27] provided a simple method for boosting the performance of machine learning models using the idea of rationales, in which human annotators would highlight a piece of text that supported their tagging conclusion. The authors created and tested a specific discriminative strategy that shows considerable improvement in sentiment categorization of movie reviews utilizing only a portion of the training set's rationales. *HateXplain*, a benchmark dataset for online hate speech detection along with rationales and target groups, was introduced by [28]. They discovered that models taught using human reasoning perform better at reducing unintended bias toward target populations.

TABLE I
EXAMPLES FROM THE *BullyExplain* DATASET, WHERE WE HAVE MARKED
THE RATIONALE TOKENS WITH GREEN HIGHLIGHTS

Tweet	Bully Label	Sentiment Label	Target
T1: Your sister is dumb , o sirf entertainment keliye Sahi hai Translation: Your sister is dumb ; she is only suitable for entertainment .	Bully	Negative	AFR
T2: saleko kon mic de diya, voice dekoh hizra jaisa hai Translation: He sounds like faggot , who gave the mic to this idiot ?	Bully	Negative	Sexual-orientation
T3: Wo har pal khubsurat ho jata hai jisme tum shamil hote ho Translation: All moments become beautiful in your presence.	Nonbully	Positive	—

After an in-depth literature review, we can conclude that most of the works on cyberbullying detection are in monolingual languages [29], [30], [31], [32], [33], [34]. Although there are some attempts to build cyberbullying detection datasets in code-mixed language, but none of them has addressed the insights on: 1) the explainability of the models; and 2) the Target group of cyberbullying victims.

III. DATASET DEVELOPMENT

This section describes the steps to construct the *BullyExplain* dataset for explainable cyberbullying detection.

A. Data Collection

First, we looked over the available code-mixed cyberbullying datasets in the literature. In Hindi–English code-mixed language, we found two cyberbullying datasets [15], [24]. Since the *BullySent* dataset [15] had already been labeled with the bully and sentiment labels, we chose it for further annotation with rationales and target class.

B. Data Annotation

1) *Annotation Training*: The annotation process was guided by three Ph.D. scholars well-versed in cyberbullying, hate speech, and offensive content. The task itself was executed by three proficient undergraduate students skilled in both Hindi and English. Initially, linguistics undergraduate students were recruited voluntarily through the department’s email list, receiving compensation in the form of gift vouchers and an honorarium. At the outset, each post in the *BullySent* dataset received annotations for the “bully” class (bully/nonbully) and the “sentiment” class (positive/neutral/negative). Focusing on the bully related tweets, we then concentrated on annotating rationales and targets. The process began with expert annotators selecting 300 samples (tweets) at random, marking words (rationales) for textual explanation and assigning appropriate target class tags. The rationale annotation method followed a similar approach to [28], where words within tweets were denoted with either a 0 or 1, indicating the absence or presence of a rationale, respectively. Target classes, including sexual-orientation, religion, attacking-relatives-and-friends (AFR), community, organization, profession, and miscellaneous, were identified based on [28] and [35].

To ensure consistency, expert annotators engaged in discussions to reconcile any disparities. This collaborative effort

TABLE II
STATISTICS OF TARGET CLASSES

Religion	Gender	ARF	Organization	Community	Profession	Miscellaneous
161	166	1067	525	173	364	594

yielded a gold standard set of 300 samples with both rationale and target annotations. These samples were divided into three sets of 100, forming the basis of a three-phase training regimen. Following each training phase, expert annotators collaborated with novices to address discrepancies and refine annotation guidelines. After the third training round, the three most skilled annotators were selected to annotate the full dataset. This rigorous process upheld annotation quality and uniformity across the dataset.

2) *Main Annotation*: We conducted our primary annotation process using the open-source platform Docanno¹ hosted on a Heroku instance. Qualified annotators were granted secure accounts exclusively for annotation and progress tracking. We commenced with a modest batch of 100 samples, gradually increasing it to 500 as annotators gained proficiency. To ensure consistency, we rectified errors made in previous batches, thus maintaining annotators’ agreement. After completing each annotation batch, final rationale labels were determined through majority voting. If there was disparity among the three annotators’ selections, an expert annotator was consulted to resolve the discrepancy. Annotators were guided to annotate posts impartially, without considering demographics, religion, or other factors. For assessing interannotator agreement (IAA) and thus affirming annotation quality, we employed the Fleiss’ Kappa score [36]. The obtained IAA scores of 0.74 for RD and 0.71 for TI tasks underscored the dataset’s acceptable quality. Table I presents a selection of instances from the annotated “*BullyExplain*” dataset.

C. Dataset Statistics

The average number of highlighted words per post in the “*BullyExplain*” dataset is 4.91, with an average of 23.15 words per tweet. The “*BullyExplain*” dataset comprises a total of 6084 samples, with 3034 labeled as nonbully and the remaining 3050 as bully posts. Furthermore, there are 1536 tweets with positive emotions, 1327 tweets with neutral sentiments, and the rest exhibiting negative moods. The statistics of the target classes in the “*BullyExplain*” dataset are presented in Table II. As

¹ <https://github.com/doccano/doccano>

depicted in Table II, approximately one-third of the bullying samples (3050) can be categorized as “ARF,” with 1067 instances belonging to this category. This observation underscores the widespread nature of cyberbullying, where victims’ relatives and friends are frequently targeted.

IV. METHODOLOGY

A. Problem Formulation

In this work, we aim to solve the task of cyberbullying detection given an input text in an interpretable and explainable fashion. An explainable cyberbullying detection model should also predict rationale, target and sentiment alongside the bully label. In this way, the model should be able to provide reasoning for its decision and the results will be more interpretable. As we have four different tasks, we keep CD and RD as the main tasks and other two tasks as secondary or auxiliary.

Formally, given an input text $X = \{x_1, \dots, x_i, \dots, x_n\}$, we intend to learn these four closely related tasks simultaneously where n is the length of input sentence. We summarize all these different tasks as follows.

- 1) *Cyberbullying detection (CD)*: It is a classification task to predict the bully label (b) for input sentence where $b \in B$ and $B \in (\text{bully, nonbully})$ is set of bully classes.
- 2) *Rationale detection (RD)*: It is a sequence labeling task to assign labels to offensive words or phrases of the input sentence. Mathematically, given an input X , the output is $R = \{r_1, \dots, r_i, \dots, r_n\}$ where r_i is the label for word x_i from the label set, $R_{\text{labels}} \in (\text{bully, nonbully})$.
- 3) *Target identification (TI)*: It is a classification task to predict which community (t) is targeted by the input sentence where $t \in T$ and $T \in (\text{religion, gender, ARF, organization, community, profession, and miscellaneous})$ is the set of target classes.
- 4) *Sentiment analysis (SA)*: It is also a classification task to predict the sentiment polarity (s) of the input sentence where $s \in S$ and $S \in (\text{positive, negative, and neutral})$ is the set of sentiment classes.

B. Redefining Explainable Cyberbullying Detection Task as Text-to-Text Generation Task

Traditional methods for dealing with multiple tasks often utilize distinct, task-specific heads for each separate task. This design makes it challenging to introduce a new task without developing a specific head for it. In this research, we introduce a novel approach by turning explainable cyberbullying detection (and other related tasks) into a text-to-text generation problem. This unifies the approach in a unique way. To transform the problem into a text generation task, we initiate the process by constructing a target sequence in natural language, denoted as Y_i , corresponding to the input sentence X_i for training. This is accomplished by concatenating all the labels from the four tasks. In the context of the RD task, we exclusively consider those rs that pertain to the offensive label set within R_{Labels} , denoted as R_{Off} . In the event of an empty offensive label set, we utilize the “NONE” token to signify the absence of offensive

tokens in the generated text. As a result, the target sequence Y_i is formally represented as

$$Y_i = \{< R_{\text{Off}} > [b][t][s]\} \quad (1)$$

where $R_{\text{Off}} \in R$.

Special characters have been inserted after each task’s prediction, as seen in the equation above. This approach facilitates the extraction of precise predictions during the testing or inference phase. By representing both the input sentence and the target sequence in natural language, it becomes feasible to leverage large pretrained models designed for sequence-to-sequence tasks in tackling this particular text-to-text generation challenge. Consequently, the problem can be reformulated as follows: given an input sequence X , the aim is to generate an output sequence Y' that encompasses all the predictions outlined in the equation above. This can be achieved through a generative model described by the following equation:

$$Y' = G(X) \quad (2)$$

where G is a generation model. One of the benefits of this methodology is the flexibility it offers, allowing the addition of new tasks simply by attaching the labels of the task to the target sequence Y , or effortlessly solving any subtask.

C. Methodology

We propose *BullyGen* (Generative explainability), a unified approach to solve the task of explainable cyberbullying detection that is capable of not only detecting whether a text is cyberbully or not but also provides reasoning by predicting rationale, target group and sentiment of the text. For better understanding, we divide our approach into two steps: 1) sequence to sequence learning; and 2) correction mechanism.

1) *Sequence to Sequence Learning (Seq2Seq)*: In the preceding section, we delineated the concept of explainable CD as a text generation task. Here, the input sentence is denoted as X , the target sequence as Y , and the output sequence containing all the requisite predictions as Y' . To address this text-to-text generation challenge, a sequence-to-sequence model with two integral components—the encoder and the decoder—proves to be instrumental. In our approach, we harnessed the capabilities of well-established pretrained models, specifically BART [37] and T5 [38], as the sequence-to-sequence models of choice. The architectural configuration of the proposed BullyGen model is visually depicted in Fig. 1.

a) *BART*: BART operates on an encoder-decoder foundation utilizing the transformer model architecture, specifically designed for text generation tasks like summarization and translation. Its pretraining involves a series of denoising objectives, including but not limited to the masking of tokens, permutation of sentences, and rotation of sentences.

b) *T5*: T5, similar to BART, is built upon an encoder-decoder framework utilizing the transformer architecture, with a specific focus on tackling a variety of text-to-text generation challenges. The primary differentiation between BART and T5 can be attributed to their distinct pretraining objectives. In the pretraining phase of T5, the transformer adopts a denoising

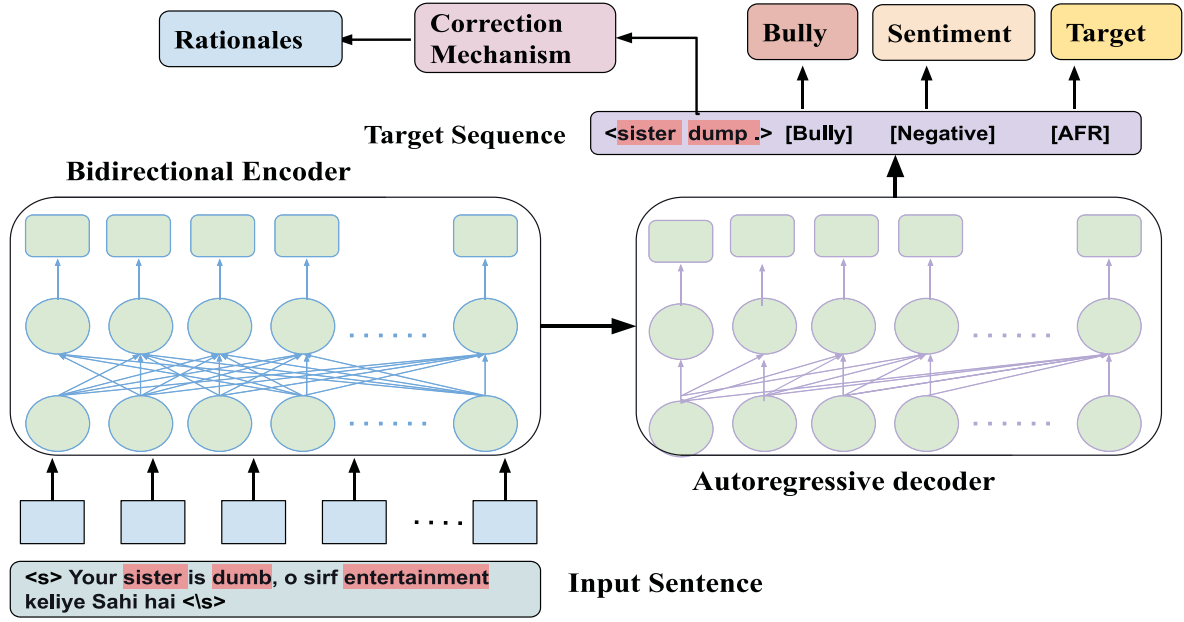


Fig. 1. BullyGen architecture.

strategy, randomly concealing 15% of the input tokens, and the decoder's task is to predict these obscured tokens. In contrast, during BART's pretraining, the decoder's role is to generate the complete input sequence.

Within the Transformer architecture, the decoder takes on the responsibility of autonomously producing the output sequence, step by step, in an autoregressive manner. At each decoding step, the model generates the subsequent token in the output sequence, guided by the information available up to that particular point. The training and inference procedures for sequence-to-sequence learning are elaborated as follows:

c) Training process: When presented with an input sentence and target sequence pair, denoted as (X, Y) , the initial stage involves supplying $X = x_0, x_1, \dots, x_i, \dots, x_n$ to the encoder module in order to acquire the concealed representation of the input, as defined in the following equation:

$$H_{EN} = G_{Encoder}(\{x_0, x_1, \dots, x_i, \dots, x_n\}) \quad (3)$$

where $G_{Encoder}$ represents encoder computations.

Upon acquiring the hidden representation, denoted as, H_{EN} , this will be coupled with all the output tokens up to the time step $t - 1$, symbolized as $Y_{<t}$. These combined elements are then channeled into the decoder module to derive the hidden state at the specific time step t , as articulated in the equation that follows:

$$H_{DEC}^t = G_{Decoder}(H_{EN}, Y_{<t}) \quad (4)$$

where $G_{Decoder}$ denotes the decoder computations.

The conditional probability for a predicted output token at the t th time step, given the input and the preceding $t - 1$ predicted tokens, is computed by applying the softmax function over the hidden state H_{DEC}^t , and it is expressed as follows:

$$P(Y'_t | X, Y_{<t}) = F_{softmax}(H_{DEC}^t W_{Gen}) \quad (5)$$

where $F_{softmax}$ represents softmax computation and W_{Gen} denotes weights of our model.

d) Training objective: For our model, we start by setting the weights, denoted by W_{Gen} , using the pretrained weights derived from established sequence-to-sequence generative models such as T5 or BART. Following this initialization, we fine-tune the model using the negative log likelihood method. In other words, we apply the maximum likelihood estimation (MLE) objective function in a supervised context to fine-tune and optimize the weights W_{Gen} , as expressed in the following equation:

$$\max_{W_{Gen}} \prod_{t=0}^T P(Y'_t | X, Y_{<t}) \quad (6)$$

MLE is a common method used to train transformer models, including language models. The goal of MLE is to determine the model parameters that optimize the likelihood of the training data. In the context of transformer models, this typically entails identifying the most suitable weights for the layers of the model, such that they maximize the probability of encountering a specific sequence of tokens in the training dataset. The loss function considers information from previous decoder time steps, encompassing the cumulative error in the model's predictions across all time steps.

e) Inference: During the training phase, we have access to both the input sentence, denoted as X , and its corresponding target sequence, denoted as Y . Consequently, we adopt the teacher forcing technique for training, which means that we use the actual target sequence instead of tokens predicted from previous time steps during the decoding process. However, during the inference phase, which is autoregressive in nature, we lack access to target sequences to guide the decoding. In this context, we replace $Y_{<t}$ with $Y'_{<t}$ in (4) to (6), where $Y'_{<t}$ signifies tokens predicted up to time step $t - 1$. To generate the predicted sequence, denoted as Y' , we employ the beam search algorithm. Beam search offers advantages over a simple greedy search

method by considering multiple alternative options, defined by the user as a hyperparameter known as beamwidth (B). This method improves the exploration for optimal tokens at each sequential stage. During beam search, the decoder simultaneously generates a group of potential output sequences, each originating from a different initial token. At each step, the decoder computes the probability distribution over the vocabulary for each candidate sequence. The process involves expanding existing sequences with the top K most probable next tokens, where K denotes the beam size. These expanded candidates are then ranked according to their cumulative probabilities, and the K sequences with the highest probabilities are retained for subsequent time steps.

2) *Correction Mechanism (CM)*: After obtaining the predicted sequence Y' , we split that sequence around special characters ($<>$, $[]$) to get the corresponding predictions for different tasks; CD, RD, TI, and SC as described in (1). Ideally, predicted elements in RD task should be the offensive words or phrases from the input sequence only. But as the sequence-to-sequence model generates target sequence from the complete vocabulary of data by which they are trained and fine-tuned on, there can be instances where the model might use a similarly different word or try to replace any code-mixed phrase with some other English language phrase. We have implemented a CM in the RD task to effectively manage code-mixed data, which often exhibits higher levels of noise, including spelling variations, abbreviations, and the absence of specific grammatical rules.

Formally, given the predicted target sequence, $Y' = \{<R'_{Off}>[b'](t')[s']\}$, every phrase in $R'_{Off} = \{r'_{off}\}$ must belong to the input sequence, X , i.e., $r'_{off} \in X$. Rather than removing predicted phrases that do not belong to X , we replace that phrase r'_{off} with its most similar phrase present in X . We use the word mover distance (WMD) [39] metric as the measure for calculating the similarity between two phrases or words. WMD is able to capture the semantic overlap between phrases by calculating the distance needed to transform one phrase to another in an embedding space. Thus, lower value of WMD implies more similarity between those phrases. We use Algorithm 1 to calculate the corrected rationales denoted by $R'_{Off_{cor}}$.

D. Experimental Settings and Hyperparameters

In this section, we elaborate on our experimental setup, elucidating the diverse hyperparameters and configurations employed in our work. All experiments were conducted on a machine equipped with an AMD EPYC 7552 48-Core Processor, 512-GB DDR4 RAM, and a suite of 5 Nvidia Ampere A100 GPUs, collectively offering 200 GB of graphics memory. We split the dataset into an 80-10-10 ratio for training, validation, and testing sets. We trained the models five times and reported the mean performance. The T5 and BART models were trained over 40 epochs, utilizing a learning rate of $2e^{-5}$ and a batch size of 16. In contrast, the baseline models were explored with various network configurations, achieving peak performance with a batch size of 32, ReLU activation function, dropout rate of 0.25, learning rate of $1e^{-4}$, and 20 epochs. We decode using

Algorithm 1 Correction Mechanism.

Input: $X = \{x\}$, $Y' = \{<R'_{Off}>[b'](t')[s']\}$
Output: $R'_{Off_{cor}} = \{r'_{off_{cor}}\}$
 $R'_{Off}, b', t', s' \leftarrow \text{split}(Y')$
 $R'_{Off_{cor}} \leftarrow \text{emptyList}$
for r'_{off} **in** R'_{Off} **do**
 if r'_{off} **in** X **then**
 $R'_{Off_{cor}} \leftarrow r'_{off}$
 end if
 if r'_{off} **not in** X **then**
 $dist \leftarrow \text{inf}$
 $WordToReplace \leftarrow \text{emptyString}$
 for x **in** X **do**
 $WMDistance \leftarrow \text{calculate WMD}(r'_{off}, x)$
 if $dist \geq WMDistance$ **then**
 $WordToReplace \leftarrow x$
 $dist \leftarrow WMDistance$
 end if
 end for
 $R'_{Off_{cor}} \leftarrow WordToReplace$
 end if
end for
Return $R'_{Off_{cor}}$

B set to 3 and top- p value set to 0.8 where top- p refers to a hyperparameter used in the sampling strategy for generating text. Top- p sampling involves choosing a subset of words whose cumulative probability exceeds a predefined threshold, p . All the models are implemented using Scikit-Learn² and pytorch³ as a backend. In our problem statement, one of the main tasks (CD) and two auxiliary tasks (TI and SA) are classification tasks. Our proposed models are evaluated on two different metrics for these tasks: 1) accuracy; and 2) F1-score. IOU F1-Score [40] based on partial matching between two span strategies has been employed for the RD task.

V. EXPERIMENTS AND RESULTS

A. Baselines Setup

We included several conventional baselines, including BiRNN, CNN-GRU, BiRNN-Attention, and BERT-finetune, as outlined in [28], for a valid comparative analysis against our proposed generative multitask model. mBERT generated sequence output has been considered as input embedding to the first three baselines. To expand the capabilities of baselines for multitasking (solving CD and RD tasks simultaneously), an extra fully connected layer is added to the shared encoded representation. The task-specific outputs are then utilized by calculating the categorical cross-entropy loss for each output. The overall loss is calculated as the sum of all individual losses.

- 1) *CNN-GRU* [41]: The input data is processed through a 1D convolutional neural network (CNN) with window

²<https://scikit-learn.org/stable/>

³<https://pytorch.org/>

TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT MULTITASK VARIANTS OF *BULLYGEN* FOR CD AND RD TASKS ON DIFFERENT EVALUATION METRICS. CD: CYBERBULLY DETECTION, RD: RATIONALE DETECTION, SA: SENTIMENT ANALYSIS, TI: TARGET IDENTIFICATION

Model	CD + RD			CD + RD + SA			CD + RD + TI			CD + RD + TI + SA		
	Bully		Rationales	Bully		Rationales	Bully		Rationales	Bully		Rationales
	ACC	F1 Score	IOU F1	ACC	F1 Score	IOU F1	ACC	F1 Score	IOU F1	ACC	F1 Score	IOU F1
BullyGen _{Bart} -base	81.41	81.23	20.02	81.24	81.22	20.82	81.61	81.53	21.10	81.45	81.44	20.40
BullyGen _{Bart} -large	81.72	81.71	21.43	81.90	81.87	20.75	82.15	81.87	19.42	80.32	80.24	20.32
Average—BullyGen _{Bart}	81.56	81.47	20.72	81.57	81.54	20.78	81.88	81.70	20.26	80.88	80.84	20.36
BullyGen _{mT5} -small	81.73	81.62	20.78	81.19	81.12	20.93	81.94	81.93	21.28	81.33	81.21	22.78
BullyGen _{mT5} -base	82.13	81.97	21.34	81.77	81.45	21.79	81.97	81.88	22.96	81.44	81.37	23.82
BullyGen _{mT5} -large	81.17	81.14	21.85	80.63	80.45	19.78	79.83	79.81	21.64	80.25	80.22	21.87
Average—BullyGen _{mT5}	81.67	81.57	21.32	81.19	81.01	20.83	81.24	81.20	21.96	80.01	80.93	20.82
BullyGen _{T5} -small	82.35	82.34	21.78	81.76	81.58	21.22	82.27	82.25	22.84	80.90	80.90	22.97
BullyGen _{T5} -base	81.67	81.59	22.35	82.33	82.33	22.38	82.36	82.19	24.68	82.95	82.96	25.17
BullyGen _{T5} -large	80.42	80.41	22.14	79.85	79.58	19.48	80.22	80.15	22.41	80.32	80.31	21.86
Average—BullyGen _{T5} -large	81.48	81.44	22.09	81.31	81.16	21.02	81.61	81.53	23.31	81.39	81.39	23.33

sizes of 2, 3, and 4, utilizing 100 filters for each window size. Subsequently, a gated recurrent unit (GRU) layer is applied in the recurrent neural network (RNN) component, followed by max pooling of the output representation derived from the hidden layers within the GRU architecture. This hidden layer is then propagated through a fully connected layer to generate the prediction logits.

- 2) *BiRNN* [42]: We pass input text through BiRNN followed by a dense layer to obtain a shared representation of the textual feature. This shared representation is then passed through task-specific layers.
- 3) *BiRNN-attention*: We pass input text to BiRNN followed by the attention layer [43]. Attended features of the text are passed through a dense layer to obtain shared representation which will be further passed through different multitask channels.
- 4) *BERT-finetune*: BERT's [44] pooled output with dimension 768 was fed to a softmax output layer.
- 5) *BullyWords-detection*: This baseline employs a rule-based word-matching approach, utilizing a predetermined set of bully words. When a sentence contains at least one word from this predefined list, it is identified as potentially having bullying content.

B. Different Multitask Variants

We have four primary tasks, and we create different multitask variations based on these tasks. CD and RD are the main tasks in each multitask variant, and we add additional secondary tasks in various combinations. We have two three-task settings: CD + RD + SA and CD + RD + TI. Additionally, there's a four-task setting, CD + RD + TI + SA. In our experiments, we test various versions of BART and T5 models, including both base and large variants of BART and all base, small, and large variants of T5 models. Given that our dataset involves a mix of two languages, we also employ all variants of mT5, a multilingual version of T5 pretrained on 101 languages.

It is crucial to emphasize that the *BullyGen* model is versatile and applicable in both single-task and multitask scenarios. In a single-task context, the target sequence comprises tokens relevant to the specific task under consideration. In contrast, in multitask scenarios, the target sequence is formed by concatenating all labels (tokens), with each label corresponding to a distinct task.

TABLE IV
RESULTS OF DIFFERENT BASELINE METHODS EVALUATED ON *BULLYEXPLAIN* DATA FOR CD AND RD TASKS

Model	CD		CD + RD		
	ACC	F1 Score	ACC	F1 Score	IOU F1 Score
BiRNN	78.62	78.51	79.43	79.53	19.62
BiRNN + Attention	79.53	79.44	79.97	80.05	19.75
CNN GRU	78.74	78.33	79.23	79.11	19.92
BERT-finetune	80.11	80.09	80.42	80.31	20.11

TABLE V
SINGLE TASK RESULTS OF *BULLYGEN* ON SA, CD, RD, AND TI TASKS

Model	CD		TI		SA		RD
	ACC	F1 Score	ACC	F1 Score	ACC	F1 Score	IOU F1
BullyGen _{Bart} -base	80.72	80.71	59.31	55.81	72.17	72.12	20.09
BullyGen _{Bart} -large	81.35	81.32	60.13	56.34	72.11	72.07	19.97
Average—BullyGen _{Bart}	81.03	81.01	59.72	56.07	72.14	72.09	20.03
BullyGen _{mT5} -small	81.52	81.52	57.53	54.29	71.95	71.58	21.43
BullyGen _{mT5} -base	81.62	81.60	58.12	55.52	72.13	72.11	21.84
BullyGen _{mT5} -large	80.12	80.12	55.02	53.19	70.56	70.82	20.35
Average—BullyGen _{mT5}	81.08	81.08	56.89	54.33	71.54	71.50	21.20
BullyGen _{T5} -small	81.46	81.28	57.44	54.23	72.43	72.22	21.13
BullyGen _{T5} -base	81.92	81.68	58.22	55.71	71.25	70.72	21.73
BullyGen _{T5} -large	80.14	80.06	57.89	55.12	70.73	70.71	20.14
Average—BullyGen _{T5}	81.17	81.01	57.85	55.02	71.47	71.21	21.00

C. Findings From Experiments

Table III shows the results of CD and RD (main tasks) on all variants of our proposed model, *BullyGen*. Other baseline results are shown in Table IV. Single task results with different variants are shown in Table V. We have also shown the effect of CM on RD task (Table VI). From all these reported results, we can conclude the following.

- 1) From Table III, we can notice that both BART and T5 variants of *BullyGen* perform equally well in solving the CD task with an average accuracy of 81.56% and 81.48% in the CD + RD multitask variant, respectively. Here, average refers to the mean performance of different versions of BART (BART-Base and BART-Large) and T5 (T5-small, T5-base, and T5-large) models within the *BullyGen* framework in Table III. The same trend is also followed in the rest of the multitask variants.
- 2) It is also clear from Table III that T5 based *BullyGen* models consistently perform better than BART based *BullyGen* models on solving the RD task. T5 based *BullyGen* outperforms the BART based *BullyGen* models with an average IOUFI score of 3.05% in the CD + RD + TI settings. The reason for this can be attributed to the fact that T5 model has been pretrained on a variety of tasks as compared to BART.

TABLE VI
ABLATION STUDY SHOWING THE EFFECT OF CORRECTION MECHANISM (CM)
ON RD TASK ON DIFFERENT MULTITASK VARIANTS;—CM: WITHOUT
CORRECTION MECHANISM

Model	CD + RD IOUF1	CD + RD + SA IOUF1	CD + RD + TI IOUF1	CD + RD SA + TI IOUF1
BullyGen _{Bart} —base	20.02	20.82	21.10	20.40
BullyGen _{Bart} —base—CM	19.40	20.10	21.02	20.20
BullyGen _{Bart} —large	21.43	20.75	19.42	20.32
BullyGen _{Bart} —large—CM	21.13	20.33	19.21	20.14
BullyGen _{T5} —small	21.78	21.22	22.84	22.97
BullyGen _{T5} —small—CM	21.05	20.82	22.07	22.11
BullyGen _{T5} —base	22.35	22.38	24.68	25.17
BullyGen _{T5} —base—CM	21.42	22.11	23.75	24.22
BullyGen _{T5} —large	22.14	19.48	22.41	21.86
BullyGen _{T5} —large—CM	21.52	19.02	21.53	21.11
BullyGen _{mT5} —small	20.76	20.93	21.28	22.78
BullyGen _{mT5} —small—CM	19.85	20.14	21.12	22.32
BullyGen _{mT5} —base	21.34	21.79	22.96	23.82
BullyGen _{mT5} —base—CM	20.72	21.25	22.22	23.05
BullyGen _{mT5} —large	21.85	19.78	21.64	21.87
BullyGen _{mT5} —large—CM	21.52	19.54	21.12	21.23

- 3) Out of all these T5 variants, BullyGen_{T5}—base performs the best in both CD and RD tasks in all the multitask variants outperforming BullyGen_{T5}—large by a significant margin. One possible reason for this under-performance of BullyGen_{T5}—large can be attributed to the size of the dataset. Due to this small size of data, BullyGen_{T5}—large may suffer from overfitting problem. This illustrates that in low resource language where the size of dataset is usually less, we should use T5-base and T5-small models.
- 4) We can also observe that T5-based variants tend to outperform the mT5 variants of *BullyGen*, indicating that pretraining language models on different languages may not necessarily lead to improved performance in code-mixed settings
- 5) From Tables III and IV, we can observe that the best proposed *BullyGen* model outperforms all the baselines significantly for solving the CD task, improving 2.53% accuracy over the best baseline, BERT-finetune. For the RD task, the improvement in IOU F1 score is 5.06%.
- 6) The observed improvements in the RD task for our study might be associated with employing T5 and BART, which are generative models, as opposed to the baselines that are discriminative models. However, it is important to note that this may be specific to this particular task and should not be generalized as a consistent trend across all tasks. While discriminative models establish boundaries within the data space, generative models aim to characterize the distribution and placement of data across the entire space, which constitutes the fundamental objective of the RD task.
- 7) Table V reports the single task results with *BullyGen*. It can be observed that even single task variants are performing at par with the best baseline model (BERT model, Table IV) showing the superiority of pretrained sequence to sequence language models.
- 8) From Tables IV and V, we can observe that out of four different multitask variants, (CD + RD + TI + SA) setting achieved the best performance for all the tasks and outperformed baselines and single task variants by

TABLE VII
EXPERIMENTAL RESULTS OF BASELINES AND DIFFERENT MULTITASK
VARIANTS OF *BULLYGEN* FOR TI AND SA TASKS ON DIFFERENT EVALUATION
METRICS

Model	CD + RD + TI Target		CD + RD + SA Sentiment		CD + RD + SA + TI Target Sentiment			
	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Baselines								
BiRNN	51.76	48.53	65.72	65.61	51.91	49.09	65.92	65.91
BiRNN + attention	51.92	48.83	65.88	65.89	52.08	49.47	66.72	66.63
CNN GRU	52.33	49.16	66.71	66.61	52.93	49.91	66.26	66.21
BERT-finetune	54.81	49.98	68.23	68.03	55.34	50.44	69.91	69.88
BullyGen								
BullyGen _{BART} -base	59.02	55.77	72.13	72.08	59.43	56.19	73.19	73.22
BullyGen _{mT5} -base	59.34	56.11	72.22	72.41	60.22	56.51	74.42	74.18
BullyGen _{T5} -base	60.72	56.78	73.15	73.88	61.77	57.47	74.81	74.72

a significant margin. These findings established the need for target, rationales and sentiment information to develop a more accurate cyberbullying detection model.

- 9) Using CD + RD + SA + TI multitask framework, we achieved the highest 25.17 IOU F1 score for the RD task illustrating that *BullyGen* model is able to learn the mapping between these three tasks efficiently, RD, SA, and TI.
- 10) We also showed how the CM is affecting the RD task (Table VI). When we remove the CM, there is a maximum drop of around 2.8% in IOUF1 score showing CM module is able to handle the noise of code-mixed data.
- 11) When we compare the results of TI and SA tasks, there is an improvement in multitask (Table VII) over single-task (Table V) settings. In all task combinations (CD + RD + SA + TI), the *BullyGen* T5-base model achieved an improvement of 1.13% and 2.50% in F1 score for TI and SA tasks, respectively. This finding establishes the efficacy of the proposed multitask model, which not only improves the performance of the main tasks (CD and RD) but also boosts the secondary tasks (TI and SA) performance.

We conducted a statistical *t*-test on five runs of both the most effective proposed model variants (BullyGen_{T5}—base in all task combinations) and the best baseline model (BERT-Finetuning), resulting in a *p*-value of 0.0002, demonstrating statistical significance with a confidence level of 95%.

D. Error Analysis

We manually reviewed a subset of test samples to assess the disparities between machine-generated rationales and bully labels compared to the judgments of human annotators. Table VIII shows the rationales detected and bully labels of few test samples obtained by different baselines and best BART and T5 based variants of the proposed model (BullyGen_{T5}—base and BullyGen_{Bart}—large). This detailed analysis of the results revealed several instances where the model faltered, some of which are discussed as follows.

- 1) For the T1 tweet, it is noticeable that the human annotator assigned the label “nonbully,” whereas all models, encompassing both baselines and *BullyGen*, predicted the label “Bully.” This prediction was driven by the identification of the offensive term “gandu” (Ass-hole), which they highlighted. This scenario illustrates the model’s challenge in comprehending the nuanced context of offensive language that isn’t targeted at any

TABLE VIII
IDENTIFIED RATIONALES FROM VARIOUS MODELS COMPARED TO THOSE BY HUMAN ANNOTATORS. TOKENS HIGHLIGHTED IN ORANGE ARE PREDICTED BY MODELS BUT NOT BY HUMAN ANNOTATORS

Model	Text	Bully Label
Human annotator (T1) Translation	Semifinal tak usi bnde ne pahochaya hai jisko tu gandu bol raha. The person you are calling asshole is the one that helped us to reach the semifinals.	Nonbully
CNN-GRU	Semifinal tak usi bnde ne pahochaya hai jisko tu gandu bol raha.	Bully
BiRNN-Attention	Semifinal tak usi bnde ne pahochaya hai jisko tu gandu bol raha.	Bully
BullyGen _{Bart-Large}	Semifinal tak usi bnde ne pahochaya hai jisko tu gandu bol raha.	Bully
BullyGen _{Bart-Large-CM}	Semifinal tak usi bnde ne pahochaya hai jisko tu gandu bol raha.	Bully
BullyGen _{T5-base}	Semifinal tak usi bnde ne pahochaya hai jisko tu gandu bol raha.	Bully
BullyGen _{T5-base-CM}	Semifinal tak usi bnde ne pahochaya hai jisko tu gandu bol raha.	Bully
Human annotator (T2) Translation	Kismat Badi kutti cheez hain... Saali Palat hi nhi rahi Luck is damned...it's not turning out.	Nonbully
CNN-GRU	Kismat Badi kutti cheez hain... Saali Palat hi nhi rahi.	Bully
BiRNN-Attention	Kismat Badi kutti cheez hain... Saali Palat hi nhi rahi.	Bully
BullyGen _{Bart-Large}	Kismat Badi kutti cheez hain... Palat hi nhi rahi.	Nonbully
BullyGen _{T5-base}	Kismat Badi kutti cheez hain... Saali Palat hi nhi rahi.	Nonbully
BullyWords-Detection	-	Bully

specific recipient. Instead, this tweet is employed more as a counterspeech. Counterspeech is a term used to describe reactions that confront or counteract hate speech, abusive language, or other harmful online communications. In the provided example, the commenter is addressing a hateful remark and encouraging the original poster to refrain from using offensive language. This underlines the model's limitation in comprehending indirect expressions.

- 2) Despite the presence of two vulgar words, "Kutti" and "Saali," in Tweet T2, the actual label is nonbully. Our proposed model, *BullyGen*, accurately predicted the label, while other two deep learning baselines, BNN-GRU and BiRNN-Attn, and a simple vocabulary-based BullyWords-Detection model made the incorrect prediction of bully. This analysis demonstrates the effectiveness of our proposed model in accurately detecting cyberbullying tweets, even in cases where vulgar language is used.

1) *Limitation:* Our study primarily delved into detecting explicit bullying posts, aiming to address a fundamental aspect of the issue. However, we acknowledge the significance of implicit bullying detection, which presents more intricate challenges due to its sarcastic nature.

E. Comparison With SOTA

We have compared our proposed model with the existing state-of-the-art model [15], where authors have developed a BERT-GRU with an attention-based multitask framework to solve CD and SA tasks simultaneously. From Table IX (showing the SOTA results), we conclude that our model reports an improvement of 1.46% and 1.83% in terms of F1 score and accuracy, respectively. These results signify that including RD tasks has the following two benefits 1) improves the performance of the main task; and 2) generates explainability along with the classification decision, making the model more interpretable.

We created a simple baseline *BullyWords-Detection* that detects whether a sentence is a bully or not based on the detection of selected vulgar or profane words. It can be seen that our

TABLE IX
RESULTS OF STATE-OF-THE-ART MODEL AND THE PROPOSED MODEL FOR CD; ST: SINGLE TASK, MT: MULTITASK (CD + SA)

Model	Accuracy	F1-score
BullyWords-Detection	52.01	51.99
ST + BERT [15]	76.97	76.89
ST + VecMap [15]	78.52	77.63
MT + BERT [15]	78.75	78.28
MT + VecMap [15]	79.60	79.65
ST-BERT + VecMap [15]	79.97	80.13
MT-BERT + VecMap [15]	81.12	81.50
BullyGen_{T5-base} (Ours)	82.95	82.96

proposed model outperforms the baseline by a significant margin showing that cyberbullying detection task is not as simple as just screening the sentence with selected sensitive words.

VI. CONCLUSION AND FUTURE WORKS

This study presents an effort to address cyberbullying detection in a code-mixed context, emphasizing the importance of explainability. The contributions of this research can be categorized into two main aspects: First, the creation of the initial code-mixed dataset for cyberbullying detection that prioritizes explainability. This dataset includes annotations not only for bully and sentiment labels but also for rationale/phrases used in decision-making and target labels. Second, the incorporation of an integrated generative framework, designed to address four concurrent tasks: CD, SA, RD, and TI. This study illustrates how a multitask challenge can be redefined as a text-to-text generation problem, allowing for the utilization of extensive pretrained sequence-to-sequence models in settings with limited linguistic resources. Through extensive experiments, we can conclude that Our proposed *BullyGen* model significantly outperforms all the baselines and reports an improvement of 1.83% accuracy over the state-of-the-art model, which establishes that utilizing sentiment, target and rationales information is beneficial to detect bully post in social media. To the best of

our knowledge, this is the first work where a generative model is used to solve a multitask classification problem.

In future, attempts will be made to extend explainable CD from code-mixed data in a multimodal setting considering image and text modalities. We would also like to work on incorporating sarcasm in our proposed framework, *BullyGen*.

DISCLAIMER

The content includes some offensive or inappropriate language. Nevertheless, such expressions are inevitable due to the inherent nature of the subject matter.

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